

OSM / Opto-PING Rung 00

Locked Reproduction of Null-Aware Specificity Gate v0.1

Status: PASS_LOCKED_NULL_AWARE_REPRODUCTION

Purpose

Rung 00 freezes the Rung 0N null-aware specificity gate and runs a fresh seed set without changing the model alternatives, selected complexity penalty, K thresholds, pass/fail criteria, or final-lock rule.

Synthetic software specificity/reproduction gate only. It does not analyze human EEG, PR-AYC-G physiology, cellular optical data, or biological memory mechanisms.

Frozen design

Item	Locked value
Source Rung 0N spec SHA-256	f11cc9662d5a666c10f29e11f70c88a04e377e6499e87ae0bb1f1632e884a6b8
Rung 00 reproduction seed	20260805
Selected penalty	0.04
Final lock rule	full reciprocal prediction win + generator-calibrated K significance
K thresholds	Rung 0N thresholds reused; no recalibration

Main results

Metric	Result
Total trials	410
Positive-control trials	60
Null / empty-sky trials	350
Positive full + K lock rate	0.950
Positive K significant rate	1.000
Positive K within 15%	1.000
Positive K within 10%	0.950
Positive median K_hat_scaled	0.991
Positive median K absolute error	0.030
Null false full + K lock rate	0.000
Null raw K significant rate	0.003
Null interpreted K rate	0.000
Null non-full predictive win rate	1.000

Lock decision table

criterion	value	target	pass
positive_full_plus_K_lock_rate >= 0.80	0.950	0.800	True
positive_K_significant_rate >= 0.90	1.000	0.900	True
positive_K_within_15_rate >= 0.90	1.000	0.900	True
null_false_full_plus_K_lock_rate <= 0.05	0.000	0.050	True
null_interpreted_K_rate <= 0.05	0.000	0.050	True
null_non_full_predictive_win_rate >= 0.85	1.000	0.850	True

Cross-validation winners

model	wins	total_folds	win_rate
sensory_drive_only	349	2460	0.142
eta_only	340	2460	0.138
zeta_only	324	2460	0.132
full_reciprocal	306	2460	0.124
standard_ping	299	2460	0.122
null_no_coupling	283	2460	0.115
colored_noise_alt	248	2460	0.101
generic_hidden_oscillator	175	2460	0.071
replay_only_basis	136	2460	0.055

Generator-level summary

Generator	Type	n	Med K	K sig	Full win	Final lock	Dominant
colored_noise_null	null	50	0.149	0.020	0.000	0.000	null_no_coupling
generic_hidden_null	null	50	0.108	0.000	0.000	0.000	null_no_coupling
permuted_label_null	null	50	0.059	0.000	0.000	0.000	null_no_coupling
replay_only_null	null	50	0.060	0.000	0.000	0.000	null_no_coupling
sensory_drive_only_null	null	50	0.073	0.000	0.000	0.000	replay_only_basis
standard_ping_null	null	50	0.053	0.000	0.000	0.000	eta_only
time_reversed_null	null	50	0.061	0.000	0.000	0.000	sensory_drive_only
full_reciprocal_positive	positive	60	0.991	1.000	0.950	0.950	full_reciprocal

Interpretation

Rung 0O reproduced the repaired Rung 0N null-aware gate under a fresh seed set. The simultaneous two-gate rule remains the critical protection: full-model prediction alone is not enough, and K significance alone is not enough. Final lock requires both.

The positive-control generator passed, while null/empty-sky generators produced zero final false full+K locks. This supports synthetic specificity of the gate under the locked Rung 0N calibration. It does not establish biological validity.

Boundary

Locked synthetic reproduction of null-aware specificity gate only; no biological, human EEG, microtubular, biophotonic, or OSM mechanism claim.